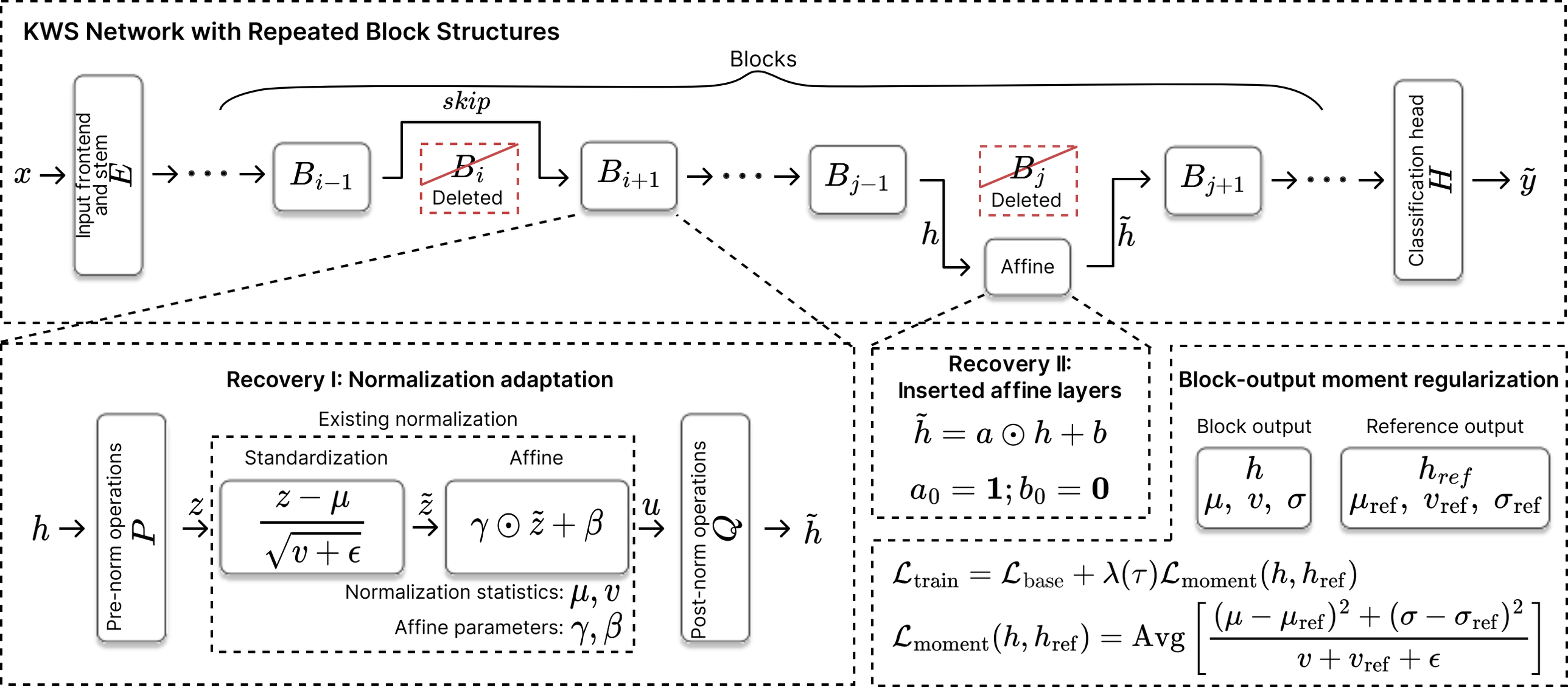

KWS Network with Repeated Block Structures
Blocks
skip
x
Input frontend and stem E
B_{i-1}
B_i
Deleted
B_{i+1}
B_{j-1}
B_j
Deleted
h
Affine
$\tilde{h}$
B_{j+1}
Classification head H
$\tilde{y}$
Recovery I: Normalization adaptation
Existing normalization
Standardization
Affine
h
Pre-norm operations P
z
$\frac{z-\mu}{\sqrt{v+\epsilon}}$
$\tilde{z}$
$\gamma \odot \tilde{z} + \beta$
u
Post-norm operations Q
$\tilde{h}$
Normalization statistics: μ, v
Affine parameters: γ, β
Recovery II: Inserted affine layers
$\tilde{h} = a \odot h + b$
$a_0 = \mathbf{1}; b_0 = \mathbf{0}$
Block-output moment regularization
Block output
Reference output
h
$\mu,\ v,\ \sigma$
h_{ref}
$\mu_{\text{ref}},\ v_{\text{ref}},\ \sigma_{\text{ref}}$
$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{base}} + \lambda(\tau)\mathcal{L}_{\text{moment}}(h, h_{\text{ref}})$
$\mathcal{L}_{\text{moment}}(h, h_{\text{ref}}) = \text{Avg}\left[\frac{(\mu-\mu_{\text{ref}})^2 + (\sigma-\sigma_{\text{ref}})^2}{v + v_{\text{ref}} + \epsilon}\right]$